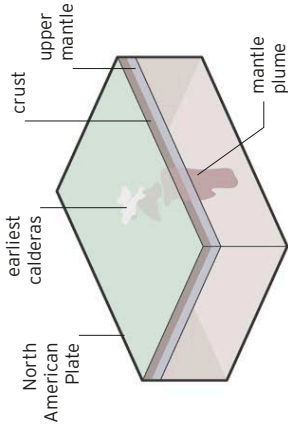


► **YELLOWSTONE CALDERA**

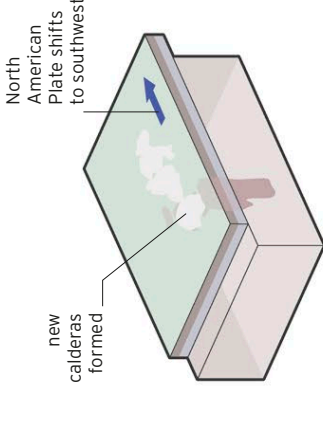
The heat source for the Yellowstone Caldera is a mantle plume—a mass of hot, semisolid rock moving slowly upward in the mantle. This supplies heat and material to the magma reservoirs in the crust under Yellowstone.

EVIDENCE OF PAST VOLCANISM

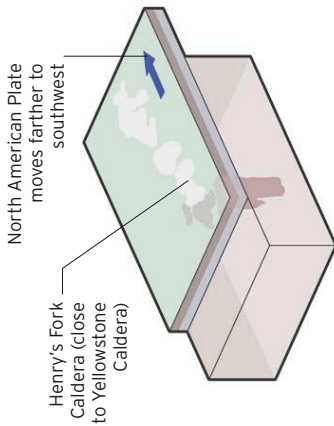
The North American Plate is moving to the southwest at the rate of about 14 miles (22 km) per million years, whereas the mantle plume currently under Yellowstone stays in approximately the same place. As a result, traces of past eruptions and calderas stretch for hundreds of kilometres to the southwest of Yellowstone.



16.5 MILLION YEARS AGO



8 MILLION YEARS AGO



1.3 MILLION YEARS AGO



Lower crustal magma reservoir
This contains hot rock and about 2 percent magma at a temperature of about 1830°F (1000°C). The amount of material here would fill the Grand Canyon over 11 times.

Dikes or vertical cracks
The upward movement of hot rock or magma occurs through dikes.



△ FUMAROLES
Around 4,000 of these holes in the ground occur within the boundaries of the Yellowstone Caldera. They release steam and other gases—some of them foul-smelling, such as hydrogen sulfide.

Mantle plume
This plume rises at least 620 miles (1,000 km) and may even originate from Earth's core-mantle boundary, which is at a depth of 1,800 miles (2,900 km).

The Yellowstone supervolcano erupts about once every 600,000–800,000 years